ModelIt!

Teacher's Guide

Can We Actually Feed 10 Billion People?

Modeling Global Agricultural Systems and Food Security

Quick Reference

Grade Level	Grade 11L1
NGSS Standard	HS-LS2-1, HS-ESS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-1, HS-ESS3-3

Use mathematical and/or computational representations to support explanations of factors that affect carrying capacity of ecosystems at different scales. Create a computational simulation to illustrate the relationships among management of natural resources, the sustainability of human populations, and biodiversity.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational models to represent the mathematical relationships between agricultural inputs, ecosystem health, and food production at global scales.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems / ESS3.C Human Impacts on Earth Systems** — Ecosystems have carrying capacities determined by resource availability. Human agricultural practices alter nutrient cycles, water systems, and biodiversity at global scales.
Crosscutting Concept	**Stability and Change** — Students analyze how agricultural systems can appear stable in the short term while undergoing long-term degradation of soil health, water resources, and biodiversity that threatens future food security.

CAST Testing Alignment

MULTIPLE CHOICE:

A country increases synthetic fertilizer application by 40% across all farmland. Based on your model, which outcome is most likely over a 20-year period: (a) sustained yield increase, (b) initial yield increase followed by decline as soil health degrades, (c) no change in yield, or (d) immediate yield decrease?

CONSTRUCTED RESPONSE:

Using evidence from your three simulation scenarios, argue for the agricultural strategy you believe is most likely to successfully feed 10 billion people by 2058 without causing irreversible environmental damage. Address both the strengths and limitations of your chosen approach.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The challenge of feeding 10 billion people is not a distant hypothetical — it is the defining agricultural challenge of the next three decades. The Green Revolution of the 1960s-70s tripled crop yields through synthetic fertilizers, irrigation, and high-yield crop varieties, averting predicted mass famines. But those gains came with hidden costs: degraded soil, depleted aquifers, polluted waterways, and collapsed biodiversity. The question now is whether we can achieve a second revolution — one that increases production while healing the land.

The Green Revolution: Triumph and Consequences

Between 1960 and 2000, global cereal grain yields tripled thanks to Norman Borlaug's high-yield wheat and rice varieties, massive expansion of irrigation, and synthetic nitrogen fertilizer (produced via the Haber-Bosch process). This saved an estimated one billion lives. However, the environmental costs have become clear: the Haber-Bosch process consumes 1-2% of global energy and produces 1.4% of CO2 emissions; nitrogen runoff has created over 400 ocean dead zones; and intensive monoculture has degraded topsoil across every continent.

Soil: The Living Foundation

Healthy topsoil is a living ecosystem containing billions of bacteria, fungi, and invertebrates per gram. These organisms decompose organic matter, cycle nutrients, retain water, and suppress plant diseases. Industrial farming practices — deep tillage, monoculture, and excessive chemical inputs — kill soil biota, reduce organic matter from 5-8% to below 2%, and accelerate erosion. It takes nature approximately 500 years to produce one inch of topsoil, but poor farming practices can destroy it in a single generation. The UN estimates that at current degradation rates, we have approximately 60 harvests of topsoil remaining.

Water: The Invisible Crisis

Agriculture consumes 70% of global freshwater withdrawals. Many of the world's most productive farming regions depend on fossil aquifers that are being depleted far faster than they recharge. The Ogallala Aquifer under the US Great Plains has declined by over 30% in some areas; India's Punjab region has dropped water tables by 10 meters in two decades. Climate change is intensifying droughts in already water-stressed regions. Without dramatic improvements in irrigation efficiency and crop water use, water scarcity may limit food production before land or nutrients do.

The Path Forward: Sustainable Intensification

Sustainable intensification aims to produce more food from existing farmland while reducing environmental impact. Key strategies include: precision agriculture using GPS and sensors to apply fertilizer and water exactly where needed (reducing waste by 20-40%); no-till farming that preserves soil structure and organic matter; crop rotation and cover cropping that restore soil health and break pest cycles; integrated pest management that reduces pesticide use by 50-70%; and genetic improvement of crop varieties for drought tolerance and nutrient efficiency. The challenge is scaling these practices globally while ensuring they're accessible to the 500 million smallholder farmers who produce 80% of the developing world's food.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify arable land area, fertilizer input, water availability, crop yield per hectare, soil health index, and total food production as the key components of the global food system.

E **Establish Relationships**

Students determine that arable land, fertilizer, and water are external inputs that farmers and policymakers control, while crop yield, soil health, and total food production are internal outcomes driven by the interaction of inputs, biology, and time.

V **Visualize & Model**

Students build a computational model showing how agricultural inputs interact with soil health and ecosystem dynamics to determine food production capacity over multiple decades.

E **Evaluate Outcomes**

Students run green revolution intensification, land expansion, and sustainable intensification scenarios to compare short-term food production gains against long-term soil degradation and environmental costs.

R **Revise & Extend**

Students add climate change impact, diet composition, or food distribution equity to explore how these factors reshape the food security challenge beyond pure production.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"By the time you're 50, there will be 10 billion people on this planet. Right now, we already use half the habitable land on Earth for farming. Where does the food come from?"*

Watch For

Show a time-lapse visualization of global population growth from 1 billion to 10 billion. Ask students: Do you think we can feed everyone?

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling the biggest systems engineering challenge on the planet — growing enough food for everyone without destroying the soil, water, and ecosystems we depend on."*

Watch For

Have students read objectives. Pre-teach 'carrying capacity' and 'soil degradation' with visual examples of healthy vs. degraded farmland.

Slide 3: Big Question

3 min

Say: *"Here's the paradox: the world already produces enough food for 10 billion people — but 735 million are hungry. So is the problem production, or something else entirely?"*

Watch For

Quick-write: Students hypothesize why hunger exists despite sufficient global production. Share out and record ideas.

Slide 4: LEVER Framework

2 min

Say: *"We're building a model that simulates 30 years of global farming — inputs, outputs, and the hidden cost of pushing land too hard for too long."*

Watch For

Preview LEVER steps. Emphasize that this model runs over DECADES, so short-term gains and long-term consequences will become visible.

Slide 5: Activity 1: Components

8 min

Say: *"What does it take to grow food at a global scale? Which factors can governments and farmers control, and which are determined by nature?"*

Watch For

Guide sorting: Arable Land Area, Fertilizer Input, and Water Availability are external (policy and farmer decisions). Yield, Soil Health, and Production are internal outcomes.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the hidden trap: fertilizer boosts yield NOW but degrades soil over YEARS. More land means less forest. Trace every connection and find the trade-offs."*

Watch For

Students map relationships, paying special attention to delayed negative feedback (fertilizer-soil loop). Discuss: Why don't farmers see the damage immediately?

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test three strategies for feeding 10 billion people: push harder with chemicals, convert more forest to farmland, or farm smarter with sustainable practices. Which one works?"*

Watch For

Students predict 30-year outcomes for each strategy. Run simulations. Compare: Which strategy feeds the most people in 2058 without crashing the system?

Slide 8: Discoveries

5 min

Say: *"The model shows the same thing real data shows: you can't cheat nature. Push the soil too hard and it pushes back. But sustainable intensification plus waste reduction can close the gap."*

Watch For

Show real-world data on yield plateaus and soil degradation trends that match model predictions. Discuss: What does this mean for policy?

Slide 9: STEM Challenge

12 min

Say: *"The UN just hired your team to design the 2058 food plan. 10 billion people, limited land, changing climate. Your plan has to work for 30 years, not just next harvest."*

Watch For

Groups design agricultural strategies specifying inputs, practices, and waste reduction targets. Present plans with model evidence for long-term viability.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Arable Land Area, Fertilizer Input, Water Availability
- **INTERNAL:** Crop Yield per Hectare, Soil Health Index, Total Food Production

Relationships

- **Fertilizer Input to Crop Yield per Hectare:** POSITIVE (+) SHORT TERM — Synthetic fertilizer provides essential nitrogen, phosphorus, and potassium that boost plant growth and grain production. However, this positive relationship has diminishing returns — beyond optimal levels, additional fertilizer provides little yield benefit while accelerating soil degradation.
- **Fertilizer Input to Soil Health Index:** NEGATIVE (-) LONG TERM — Excessive synthetic fertilizer kills beneficial soil microorganisms, reduces organic matter content, acidifies topsoil, and makes crops dependent on chemical inputs. Over decades, this degrades the soil's natural fertility, creating a dependency cycle where more fertilizer is needed just to maintain previous yield levels.
- **Soil Health Index to Crop Yield per Hectare:** POSITIVE (+) — Healthy soil with rich organic matter, diverse microbial communities, and good structure provides natural nutrient cycling, water retention, and disease suppression that support strong crop yields. As soil health declines, yields become increasingly dependent on external chemical inputs rather than the soil's inherent productivity.

Key Discoveries

- Maximum fertilizer input produces impressive short-term yield gains but degrades soil organic matter, reduces microbial diversity, and increases erosion — creating a ticking time bomb where yields eventually decline despite increasing chemical inputs
- Expanding farmland into forests and grasslands provides immediate food production gains but destroys carbon sinks, collapses biodiversity, and often yields less per hectare because newly cleared land has thinner topsoil
- Sustainable intensification — moderate inputs combined with soil conservation, crop rotation, and precision agriculture — can maintain or increase yields while preserving soil health, but requires investment in technology and farmer training
- The global food system already produces enough calories for 10 billion people, but approximately 40% is lost to waste, animal feed conversion, and biofuels — reducing waste may be more effective than growing more food

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We just need to grow more food to end hunger"	The world already produces approximately 6,000 calories per person per day — more than enough for everyone. Hunger is primarily caused by poverty, distribution failures, conflict, and waste, not insufficient production. Approximately 40% of food produced globally is lost or wasted. Simply growing more food without addressing distribution and waste will not end hunger — it will just produce more waste and more environmental damage.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Organic farming can't feed the world"	While organic yields are typically 10-25% lower per hectare than conventional farming, organic practices maintain soil health, reduce pollution, and require fewer expensive inputs. Research from the Rodale Institute's 40-year trial shows that organic yields match conventional yields during droughts because healthy soil retains more water. The real answer is likely neither purely organic nor purely conventional, but an integrated approach that uses the best practices from both — sustainable intensification.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Technology alone will solve the food crisis"	Technology is essential — precision agriculture, drought-resistant crops, and improved storage can dramatically improve efficiency. But technology cannot solve a problem that is fundamentally about politics, economics, and behavior. If wealthy nations continue wasting 30-40% of food while poor nations lack refrigeration and roads, no amount of yield improvement will achieve food security. The model shows that waste reduction and equitable distribution have larger impacts on global food access than any single technology.